

Javad Noorbakhsh

Bioinformatics & Machine Learning Scientist

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Education

- **Ph.D., Physics** — Boston University, Boston, MA (September 2009 - December 2014)
- **M.Sc., Physics** — Sharif University of Technology, Tehran, Iran (September 2006 - December 2008)
- **B.Sc., Electrical Engineering** — Amirkabir University of Technology, Tehran, Iran (September 2001 - December 2005)

Experience

Bioinformatics & Machine Learning Scientist, Jackson Laboratory for Genomic Medicine, Farmington, CT July 2024 - Present

- Applied pathology foundation models to accurately identify and classify blood cell types within Giemsa-stained tissue slides.
- Engineered ML algorithms to cross-evaluate and compare spatial omics profiles between human and mouse models.
- Benchmarked the impact of H&E staining variations to optimize foundation model diagnostic performance.
- Evaluated spatial transcriptomics data from cancer patient cohorts to quantify therapeutic effects on microenvironment composition.
- Formulated mathematical models of tumor evolution and subclonal dynamics, enhancing precision in treatment strategy selection.
- Co-authored peer-reviewed reviews and competitive grant proposals advancing AI applications in spatial omics.

Principal Scientist, Kojin Therapeutics, Boston, MA April 2022 - February 2024

- Spearheaded AI-driven multi-omics analysis, directly enabling successful fundraising and a key strategic pharmaceutical partnership.
- Established high-throughput assay screening pipelines, successfully identifying novel small-molecule hits for drug development.
- Deployed graph neural networks (GNNs) for molecular design, accelerating target discovery and experimental drug candidate validation.
- Architected attention-based multi-instance learning pipelines on histology images, uncovering potential drug-sensitive cancer subtypes.

Lead Scientist, Broad Institute of MIT and Harvard, Cambridge, MA January 2020 - April 2022

- Orchestrated the Cancer Dependency Map omics team, managing end-to-end RNAseq, WES, and WGS pipelines to deliver quarterly releases across public, institutional, and commercial portals.
- Standardized and automated production pipelines across major projects, achieving highly reproducible and timely data distribution.
- Directed ML initiatives aligning patient tumor NGS data with cell lines, substantially boosting alignment accuracy and tool reliability.

Research Scientist, The Jackson Laboratory for Genomic Medicine, Farmington, CT October 2017 - December 2019

- Pioneered a cloud-based deep learning framework for tumor state classification across histopathology images, attaining AUCs > 0.9 across 19 cancer types.
- Guided the development and deployment of an image processing framework for 3D organoid segmentation, producing an open-source software package.

Postdoctoral Fellow, The Jackson Laboratory for Genomic Medicine, Farmington, CT February 2015 - October 2017

- Investigated copy number variations and mutation calling dynamics in cancer, identifying key biomarker correlations with patient survival.
- Constructed mathematical models of clonal evolution, improving quantitative assessment of genetic heterogeneity across diverse tumor types.

Skills

Quantitative Analysis: Machine Learning, Deep Learning, Bioinformatics, Statistics, Image Processing, Mathematical Modeling, Cheminformatics

Programming: Python, TensorFlow/PyTorch, Scikit-learn, R, Bash, HPC, Cloud Computing, Containerization, Kubernetes, Version Control, C++, MATLAB

Leadership: Formal management, Leadership, Mentorship, Team Collaboration

Publications

- Prediction of outcome from spatial Protein profiling of triple-negative breast cancers
Ali Foroughi Pour, Te-Chia Wu, Javad Noorbakhsh, Jan Martinek, Santhosh Sivajothi, et al. — Communications Medicine, 2026
- RANKL inhibition spatially rewires the microenvironment of luminal breast cancer
Mario Rodriguez-del Collado, Andrea Vethencourt, Alexandra Barranco, Jaime Martinez de Villarreal, David Valcárcel-Linares, et al. — medRxiv, 2026
- DIANNE: Segmentation-Free Localization of Histology Differential Attributes
Sergii Domanskyi, Jill C Rubinstein, Todd B Sheridan, Adam Thiesen, Javad Noorbakhsh, et al. — bioRxiv, 2026
- Spatial multiomic characterization of spontaneous lung tumors in aged mice.

Adam Thiesen, Sema Akkurt, Javad Noorbakhsh, Te-Chia Wu, Andrew Salner, et al. — Cancer Research, 2026

- **Phone-based portable slide digitization system for AI-enabled histopathology**
Sergii Domanskyi and Todd Sheridan and Javad Noorbakhsh and Jeffrey H Chuang and Jill Carol Rubinstein — Cancer Research, 2026
- **Emerging AI approaches for cancer spatial omics**
Javad Noorbakhsh and Ali Foroughi Pour and Jeffrey Chuang — None, 2025
- **Convolutional neural networks for classification of cancer histological images**
Jeffrey Hsu-Min Chuang and Javad Noorbakhsh and Kourosh Zarringhalam and Saman Farahmand and Mohammad Soltanieh-ha — None, 2025
- **Convolutional neural networks for classification of cancer histological images**
Jeffrey Hsu-Min Chuang and Javad Noorbakhsh and Kourosh Zarringhalam and Saman Farahmand and Mohammad Soltanieh-ha — None, 2025
- **Germline variation contributes to false negatives in CRISPR-based experiments with varying burden across ancestries**
Sean A Misek, Aaron Fultineer, Jeremie Kalfon, Javad Noorbakhsh, Isabella Boyle, et al. — Nature communications, 2024
- **The enzyme glutamate-cysteine ligase (GCL) is a target for ferroptosis induction in cancer**
John K Eaton, Priya Chatterji, Laura Furst, Sneha Basak, Ayesha M Patel, et al. — BioRxiv, 2024
- **Abstract A032: Cellos: High-throughput deconvolution of 3D organoid dynamics at cellular resolution for cancer pharmacology**
Patience Mukashyaka, Pooja Kumar, David J Mellert, Shadae Nicholas, Javad Noorbakhsh, et al. — Cancer Research, 2024
- **High-throughput deconvolution of 3D organoid dynamics at cellular resolution for cancer pharmacology with Cellos**
Patience Mukashyaka, Pooja Kumar, David J Mellert, Shadae Nicholas, Javad Noorbakhsh, et al. — Nature Communications, 2023
- **Cellos: High-throughput deconvolution of 3D organoid dynamics at cellular resolution for cancer pharmacology**
Patience Mukashyaka, Pooja Kumar, David J Mellert, Shadae Nicholas, Javad Noorbakhsh, et al. — bioRxiv, 2023
- **Simulating the phase behavior of the Kuramoto tree**
Mohammad Javad Nouhi and Javad Noorbakhsh — arXiv preprint arXiv:2306.14928, 2023
- **Defining the landscape of cancer vulnerabilities that are engendered by germline genetic variation**
Sean Misek, Aaron Fultineer, Jeremie Kalfon, Javad Noorbakhsh, Isabella Boyle, et al. — The Journal of Pharmacology and Experimental Therapeutics, 2023
- **Computational estimation of quality and clinical relevance of cancer cell lines**
Lucia Trastulla and Javad Noorbakhsh and Francisca Vazquez and James McFarland and Francesco Iorio — None, 2022
- **Associations between structural variant signatures and drug sensitivity in cell lines**
Simona Dalin, Frank Dubois, Shu Zhang, Alexander Crane, Jeremie Kalfon, et al. — Cancer Research, 2022
- **Ancestry bias in CRISPR guide design impedes discovery of genetic dependencies**

Sean Alexander Misek, Aaron Fultineer, Jeremie Kalfon, Javad Noorbakhsh, Isabella Boyle, et al. — Cancer Research, 2022

- **Bridging the gap between cancer cell line models and tumours using gene expression data**

Javad Noorbakhsh and Francisca Vazquez and James M McFarland — British Journal of Cancer, 2021

- **Abstract PO-003: Deep learning identifies conserved pan-cancer tumor features**

Javad Noorbakhsh, Saman Farahmand, Ali Foroughi pour, Sandeep Namburi, Dennis Caruana, et al. — Clinical Cancer Research, 2021

- **Intratumor heterogeneity quantification using 3D segmentation of organoids co-culture systems**

Patience Mukashyaka, Javad Noorbakhsh, Pooja Kumar, Elise Courtois, Olga Anczukow, et al. — Cancer Research, 2021

- **Deep learning-based cross-classifications reveal conserved spatial behaviors within tumor histological images**

Javad Noorbakhsh, Saman Farahmand, Sandeep Namburi, Dennis Caruana, David Rimm, et al. — Nature communications, 2020

- **Repeat expansions confer WRN dependence in microsatellite-unstable cancers**

Niek van Wietmarschen, Sriram Sridharan, William J Nathan, Anthony Tubbs, Edmond M Chan, et al. — Nature, 2020

- **Treating cancer as an invasive species**

Javad Noorbakhsh and Zi-Ming Zhao and James C Russell and Jeffrey H Chuang — None, 2020

- **Foroughi pour A, Namburi S, Caruana D, Rimm D, et al**

Javad Noorbakhsh and Saman Farahmand — Deep learning-based cross-classifications reveal conserved spatial behaviors within tumor histological images. Nat Commun, 2020

- **Ali Foroughi pour, Sandeep Namburi, Dennis Caruana, David Rimm, Mohammad Soltanieh-ha, Kourosh Zarringhalam, and Jeffrey H**

Javad Noorbakhsh and Saman Farahmand — Chuang. Deep learning-based cross-classifications reveal conserved spatial behaviors within tumor histological images. nature communications, 2020

- **Integrative deep learning for PanCancer molecular subtype classification using histopathological images and RNAseq data**

Fatima Zare and Javad Noorbakhsh and Tianyu Wang and Jeffrey H Chuang and Sheida Nabavi — None, 2020

- **AzinNet: A wavelet convolutional neural network for pathology image analysis**

Ali Foroughi Pour and Javad Noorbakhsh and Jeffrey Chuang — Cancer Research, 2020

- **Longitudinal molecular trajectories of diffuse glioma in adults**

Floris P Barthel, Kevin C Johnson, Frederick S Varn, Anzhela D Moskalik, Georgette Tanner, et al. — Nature, 2019

- **CCNE1 amplification is associated with poor prognosis in patients with triple negative breast cancer**

Zi-Ming Zhao, Susan E Yost, Katherine E Hutchinson, Sierra Min Li, Yate-Ching Yuan, et al. — BMC cancer, 2019

- **Machine learning in biology and medicine**

PhD Javad Noorbakhsh and MS Harshpreet Chandok and PhD R Krishna Murthy Karuturi and PhD Joshy George — None, 2019

- **The PDX Data Commons and Coordinating Center (PDCCC) for PDXNet in support of preclinical research**
Jacqueline Rosains, Anuj Srivastava, Wingyi Woo, Vishal Sarsani, ZiMing Zhao, et al. — Cancer Research, 2019
- **Deep learning functional associations using histopathology images**
Javad Noorbakhsh, Saman Farahmand, Mohammad Soltanieh Ha, Sandeep Namburi, Kourosh Zarringhalam, et al. — Cancer Research, 2019
- **Distribution-based measures of tumor heterogeneity are sensitive to mutation calling and lack strong clinical predictive power**
Javad Noorbakhsh and Hyunsoo Kim and Sandeep Namburi and Jeffrey H Chuang — Scientific Reports, 2018
- **High-resolution deconstruction of evolution induced by chemotherapy treatments in breast cancer xenografts**
Hyunsoo Kim, Pooja Kumar, Francesca Menghi, Javad Noorbakhsh, Eliza Cerveira, et al. — Scientific reports, 2018
- **The PDX Data Commons and Coordinating Center (PDCCC) for PDXNet**
Anurag Sethi, Anuj Srivastava, Xingyi Woo, Vishal Sarsani, Ziming Zhao, et al. — Cancer Research, 2018
- **Evolution of an intratumoral ecology susceptible to successive treatment in breast cancer xenografts**
Hyunsoo Kim, Pooja Kumar, Francesca Menghi, Javad Noorbakhsh, Eliza Cerveira, et al. — bioRxiv, 2018
- **Uncertainties in tumor allele frequencies limit power to infer evolutionary pressures**
Javad Noorbakhsh and Jeffrey H Chuang — Nature genetics, 2017
- **From intracellular signaling to population oscillations: bridging size-and time-scales in collective behavior**
Allyson E Sgro, David J Schwab, Javad Noorbakhsh, Troy Mestler, Pankaj Mehta, et al. — Molecular systems biology, 2015
- **Modeling oscillations and spiral waves in Dictyostelium populations**
Javad Noorbakhsh and David J Schwab and Allyson E Sgro and Thomas Gregor and Pankaj Mehta — Physical Review E, 2015
- **Two-dimensionality of yeast colony expansion accompanied by pattern formation**
Lin Chen, Javad Noorbakhsh, Rhys M Adams, Joseph Samaniego-Evans, Germaine Agollah, et al. — PLoS computational biology, 2014
- **Stochasticity and universal dynamics in communicating cellular populations**
Javad Noorbakhsh, Pankaj Mehta, Allyson Sgro Collaboration, David Schwab Collaboration, Troy Mestler Collaboration, et al. — APS March Meeting Abstracts, 2014
- **Intrinsic noise of microRNA-regulated genes and the ceRNA hypothesis**
Javad Noorbakhsh and Alex H Lang and Pankaj Mehta — PLoS One, 2013
- **Quantitative Analysis of How RET Receptor Activation is Coupled to ERK and Akt Signaling: 638**
Simin Li and Javad Noorbakhsh and Thomas V Riera and Pankaj Mehta and Adrian Whitty — Protein Science, 2013
- **Modeling Excitable Systems Coupled Through External Medium**
Javad Noorbakhsh and Pankaj Mehta — APS March Meeting Abstracts, 2013

- **Quantitative Model of microRNA-mRNA interaction**
Javad Noorbakhsh and Alex Lang and Pankaj Mehta — APS March Meeting Abstracts, 2012
- **The Mouseion at the JAXlibrary**
Javad Noorbakhsh and Saman Farahmand and Ali Foroughi Pour and Sandeep Namburi and Dennis Caruana — None
- **CCNE1 Amplification Is Associated with Poor Prognosis in Patients with Triple Negative**
Zi-Ming Zhao, Susan Yost, Katherine Hutchinson, Sierra Li, Yate-Ching Yuan, et al. — None
- **A comparison of gene regulation by eukaryotic microRNAs and prokaryotic sRNAs**
J Noorbakhsh and A Lang and P Mehta — None